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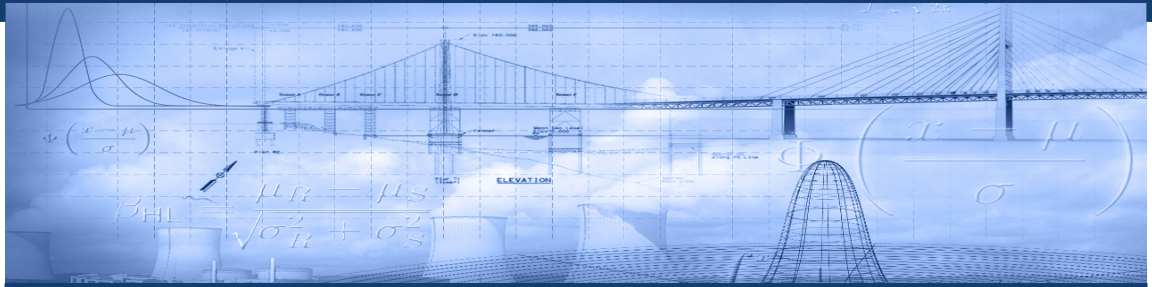


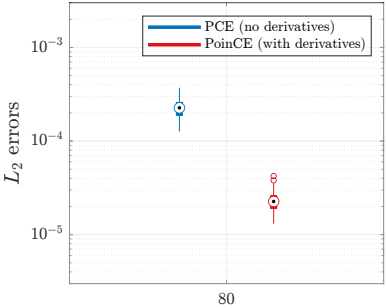
Gradient-enhanced surrogate modeling and sensitivity analysis with Poincaré chaos expansions

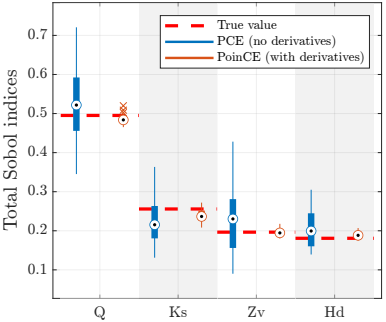
PILearnWater Chair, 2026 March 11

D. Heredia

Spoilers

- So now you know how to use perform PCE (polynomial chaos expansion) to construct surrogate models and compute Sobol indices.
- In this part we replace orthogonal polynomials by Poincaré bases.
- These bases allow to incorporate derivative information to improve the estimators.





Check all the details in [Roustant et al. (2025)]

Outline

Introduction

Poincaré basis

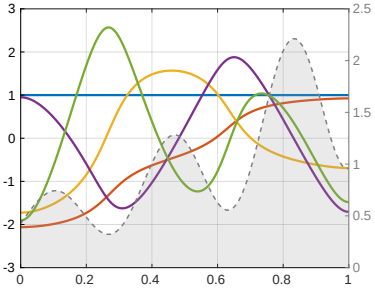
Chaos decomposition using derivatives

Numerical results

Poincaré eigenbasis (1D)

- Consider $f_X : (a, b) \to \mathbb{R}^+$ a 1D density function.
- The Poincaré basis $(\psi_n)_{n\geq 0}$ is the only orthogonal basis of $L^2_{f_X}$ such that $\psi_0 \equiv 1$ and whose family of derivatives $(\psi'_n)_{n\geq 1}$ is itself an orthogonal basis:

$$\langle \psi_n, \psi_m \rangle_{f_X} = 0, \quad \langle \psi'_n, \psi'_m \rangle_{f_X} = 0, \quad \text{for all } n \neq m.$$



Examples:

- Uniform distribution $\mathcal{U}(0, 1)$:

$$\psi_n(x) = \cos(n\pi x), \quad \psi'_n(x) = -\, n \sin(n\pi x).$$

- Normal distribution $\mathcal{N}(0, 1)$ – Hermite polynomials:

$$H_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}, \quad H'_n(x) = nH_{n-1}(x).$$

Weighted Poincaré eigenbasis (1D)

- Now consider a weight $w\colon (a,b) \to \mathbb{R}^+$.
- The weighted Poincaré basis $(\psi_n)_{n\geq 0}$ is the only orthogonal basis such that $\psi_0 \equiv 1$ and whose family of derivatives $(\psi_n')_{n\geq 1}$ is an orthogonal basis of $L^2_{w\,f_X}$:

$$\langle \psi_n, \psi_m \rangle_{f_X} = 0, \quad \langle \psi_n', \psi_m' \rangle_{w\,f_X} = \langle w\,\psi_n', \psi_m' \rangle_{f_X} = 0, \quad \text{for all } n \neq m.$$

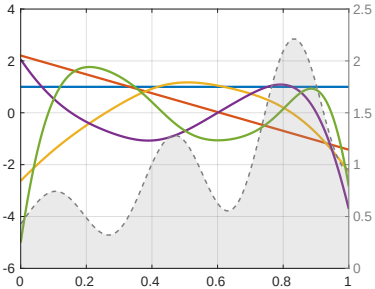
- The choice of the weight w modifies the whole basis $(\psi_n)_{n\geq 0}$.
  When $w \equiv 1$ we obtain the classical Poincaré basis.

The linear standard weight w_{lin}

- Here we consider the weight $w = w_{\text{lin}}$, which enforces $\psi_1(x)$ to be linear.
- It is defined as

$$w_{\text{lin}}(x) = -\frac{1}{f_X(x)} \int_a^x (y-m)\ f_X(y)\, dy, \quad \text{where} \quad m = \int_a^b y\, f_X(y)\, dy$$

  (we explain this later).
- It is also called the Stein kernel.



Examples:

- Normal distribution $\mathcal{N}(0,1)$, $w_{\text{lin}}(x)$=1
  Hermite polynomials.
- Gamma distribution Gam(α), $w_{\text{lin}}(x) = x$
  Laguerre polynomials.
- Beta distribution $B(a,b)$, $w_{\text{lin}}(x) = \frac{1}{\alpha+\beta} x(1-x)$
  Jacobi polynomials.

⚠ Poincaré bases match with polynomials in these three cases only.

Proper definition (ugly maths)

- Consider the Sturm-Liouville operator

$$L_w g = \frac{1}{f_X}\left(w\, g'\, f_X\right)'.$$

- The key property: if g belongs to

$$\mathcal{C}_N(a,b) = \{g \in \mathcal{C}(a,b) \,|\, (w\, g' f_X)(a) = (w\, g' f_X)(b) = 0\} \quad \text{(Neumann conditions)},$$

then

$$\langle (-Lg), h\rangle_{f_X} = \langle w\, g', h'\rangle_{f_X}, \quad \text{for all } h.$$

- Indeed:

$$\langle (-L_w g), h\rangle_{f_X} = -\int_a^b \frac{1}{f_X}\left(w\, g'\, f_X\right)' h\, f_X = -\int_a^b \left(w\, g'\, f_X\right)' h$$

$$= -\left[(w\, g'\, f_X)\, h\right]\Big|_a^b + \int_a^b w\, g' h'\, f_X = \langle w\, g', h'\rangle_{f_X}$$

Proper definition (ugly maths)

- Under some conditions on f_X and w, the spectrum of $-L_w$ is discrete:
  There is a family $(\psi_n)_{n\geq 0}$ of eigenfunctions of $-L_w$ in $\mathcal{C}_N(a,b)$. That is

$$\begin{cases} -L_w\,\psi_n = \lambda_n\,\psi_n, & \text{with } \lambda_n \in \mathbb{R}, \\ (w\,\psi_n'\,f_X)'(a) = (w\,\psi_n'\,f_X)'(b) = 0. \end{cases}$$

- The family $(\psi_n)_{k\geq 0}$ forms an orthogonal basis of $L^2_{f_X}$.

Claim: $(\psi_n)_{n\geq 0}$ is the Poincaré basis.

- Indeed, for all h we have

$$\lambda_n \langle \psi_n, h\rangle_{f_X} = \langle -L_w\psi_n, h\rangle_{f_X} = \langle w\,\psi_n', h'\rangle_{f_X}.$$

- In particular

$$\langle w\,\psi_n', \psi_n'\rangle_{f_X} = \underbrace{\lambda_n}_{\geq 0}\langle \psi_n, \psi_n\rangle_{f_X}, \quad \langle w\,\psi_n', \psi_m'\rangle_{f_X} = \lambda_n\langle \psi_n, \psi_m\rangle_{f_X} = 0, \quad \text{for all } n \neq m$$

  $\Rightarrow (\psi_n')_{k\geq 1}$ is orthogonal in $L^2_{wf_X}$. One can show that it is a dense system.

- Also,

$$-L1 = 0 = \underbrace{0}_{\lambda_0} \times \underbrace{1}_{\psi_0}.$$

Examples

The Poincaré basis functions are solutions of

$$\begin{cases} -L_w\, g = -\dfrac{1}{f_X}(w\, g'\, f_X)' = \lambda\, g, \qquad \text{with } \lambda \geq 0, \\ (w\, g'\, f_X)'(a) = (w\, g'\, f_X)'(b) = 0. \end{cases}$$

- Uniform distribution $\mathcal{U}(0,1)$ with $w \equiv 1$:

$$\begin{cases} -L_w\, g = -g'' = \lambda\, g, \\ g'(0) = g'(1) = 0. \end{cases}$$

$$\Leftrightarrow$$

$$\begin{cases} g(x) = A\, \cos(\sqrt{\lambda}\, x) + B\, \sin(\sqrt{\lambda}\, x)\,, \\ B\sqrt{\lambda} = -A\sqrt{\lambda}\sin(\sqrt{\lambda}) = 0. \end{cases}$$

$$\Leftrightarrow$$

$$\sqrt{\lambda} = \pi n, \quad g(x) = A\underbrace{\sin(\pi n\, x)}_{\psi_n(x)}, \quad n \geq 0.$$

Examples: the three cases where the Poincaré basis is polynomial

The Poincaré basis functions are solutions of

$$\begin{cases} -L_w\, g = -\dfrac{1}{f_X}(w\, g'\, f_X)' = \lambda\, g, \qquad \text{with } \lambda \geq 0, \\ (w\, g'\, f_X)'(a) = (w\, g'\, f_X)'(b) = 0. \end{cases}$$

- Normal distribution $\mathcal{N}(0,1)$ with $w_{\text{lin}}(x) = 1$

$$L_w\, g(x) = g''(x) - x\, g'(x), \quad \text{(Hermite operator).}$$

- Gamma distribution Gam(α) ($f_X(x) \sim x^{\alpha-1}e^{-x}$), with $w_{\text{lin}}(x) = x$

$$L_w\, g(x) = x g''(x) + (\alpha - x) g'(x) \quad \text{(Laguerre operator).}$$

- Beta distribution $B(\alpha,\beta)$ ($f_X(x) \sim x^{\alpha-1}(1-x)^{\beta-1}$) with $w_{\text{lin}}(x) = \frac{1}{\alpha+\beta}x(1-x)$

$$L_w\, g(x) = \frac{1}{\alpha+\beta}x(1-x)g''(x) + \frac{1}{\alpha+\beta}(\alpha\,(1-x) - \beta\, x)g'(x) \quad \text{(Jacobi operator).}$$

Basis numerical estimation

- The Poincaré basis functions are solutions of

$$\begin{cases} -L_w\, g = -\dfrac{1}{f_X}(w\, g'\, f_X)' = \lambda\, g, & \text{with } \lambda \geq 0, \\ (w\, g'\, f_X)'(a) = (w\, g'\, f_X)'(b) = 0. \end{cases}$$

- Equivalently, they are solution of

$$\lambda \langle g, h \rangle_{f_X} = \langle w\, g', h' \rangle_{f_X}, \quad \lambda \geq 0, \quad \text{for all } h.$$

- The couples (ψ_n, λ_n) can be approximated, for instance, using finite elements.

We have to talk about w_{lin}...

- Consider

$$w_{\text{lin}}(x) = -\frac{1}{f_X(x)} \int_a^x (y-m)\ f_X(y)\, dy, \quad \text{where} \quad m = \int_a^b y\, f_X(y)\, dy.$$

- If $g(x) = x - m$, we have

$$-L_{w_{\text{lin}}} g(x) = \frac{1}{f_X}(w_{\text{lin}}\, g' f_X)'(x) = \frac{1}{f_X(x)} \left(\int_a^x (y-m)\ f_X(y)\, dy \right)'(x) = x - m = g(x).$$

- Since ψ_1 is the only monotonic eigenfunction, we have $\psi_1 = g$.

Numerical approximation: solve

$$\begin{cases} (w_{\text{lin}}\, f_X)'(x) = -(x-m) f_X(x), & x \in (a,b), \\ (w_{\text{lin}}\, f_X)(a) = 0, \end{cases}$$

then recover $w_{\text{lin}} = (w_{\text{lin}}\, f_X)/f_X$.

- Bonus: For any other monotonic centered function $g \in L^2_{f_X}$, there exists an associated weight w_g for which $\psi_1 = g$.

Why is it called Poincaré basis?

Fun fact: the measure with density f_X satisfies the weighted Poincaré inequality

$$\mathrm{Var}_{f_X}(g) := \int_a^b g^2\, f_X - \left(\int_a^b g\, f_X\right)^2 \leq \underbrace{C_P(f_X, w)}_{1/\lambda_1} \int_a^b w\,(g')^2\, f_X, \quad \text{for all } g$$

(please come to see tomorrow's training session).

Application to GSA: if the measure of X_i satisfies a weighted Poincaré inequality with weight w_i, then

$$S_i^{\top} \leq \frac{C_P(f_{X_i}, w_i)}{\mathrm{Var}(f(X))}\, \underbrace{\mathbb{E}\left[w_i(X_i)\left(\frac{\partial f}{\partial x_i}(\boldsymbol{X})\right)^2\right]}_{\text{Weighted } DGSM}.$$

$\Rightarrow$ Useful to perform screening.

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Poincaré expansions and derivatives

Recall: for any weight w, the associated $(\psi_n)_{n\geq 0}$ and $(\psi'_n)_{n\geq 1}$ are orthogonal bases.

Recall the notation: $\psi_{\boldsymbol{\alpha}}(\boldsymbol{x}) = \psi_{\alpha_1}(x_1)\dots\psi_{\alpha_d}(x_d)$. We have

PoinCE

$$\mathcal{M}(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}\in\mathcal{A}} c_{\boldsymbol{\alpha}}\psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$$

$$\xrightarrow{\frac{\partial}{\partial x_i}}$$

PoinCE-der(i)

$$\frac{\partial}{\partial x_i}\mathcal{M}(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}\in\mathcal{A}} c_{\boldsymbol{\alpha}}\frac{\partial}{\partial x_i}\psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$$

- Same idea as PCE, just a different basis

- Lets us estimate a subset of $c_{\boldsymbol{\alpha}}$ (since basis functions which are constant in x_i vanish by differentiation)

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PoinCE-der(i)

$$\frac{\partial}{\partial x_i}\mathcal{M}(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}\in\mathcal{A},\alpha_i>0} c_{\boldsymbol{\alpha}}\frac{\partial}{\partial x_i}\psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$$

- Same idea as PCE, just a different basis

- Lets us estimate a subset of $c_{\boldsymbol{\alpha}}$ (since basis functions which are constant in x_i vanish by differentiation)

Combined regression using evaluations and derivatives at once

Adcock & Sui (2019)

Regression problems

$$\mathcal{M}(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}\in\mathcal{A}} c_{\boldsymbol{\alpha}} \psi_{\boldsymbol{\alpha}}(\boldsymbol{x}) \qquad \Rightarrow \qquad \boldsymbol{\Psi}\boldsymbol{c} \approx \boldsymbol{y}$$

$$\sqrt{w_k(x_k)}\frac{\partial \mathcal{M}}{\partial x_k}(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}\in\mathcal{A}} c_{\boldsymbol{\alpha}} \sqrt{w_k(x_k)}\,\frac{\partial \psi_{\boldsymbol{\alpha}}}{\partial x_k}(\boldsymbol{x}) \qquad \Rightarrow \qquad \boldsymbol{\Psi}_{\partial_k}\boldsymbol{c} \approx \boldsymbol{y}_{\partial_k}$$

where

$$\Psi_{i\,\alpha} = \psi_\alpha(\boldsymbol{x}^{(i)}),\ y_i = \mathcal{M}(\boldsymbol{x}^{(i)}), \quad (\boldsymbol{\Psi}_{\partial_k})_{i\,\alpha} = \sqrt{w_k(x_k^{(i)})}\frac{\partial}{\partial x_k}\psi_\alpha(\boldsymbol{x}^{(i)}),\ (y_{\partial_k})_i = \sqrt{w_k(x_k^{(i)})}\mathcal{M}(\boldsymbol{x}^{(i)}).$$

Combined regression

$$\begin{bmatrix} \boldsymbol{\Psi} \\ \boldsymbol{\Psi}_{\partial_1} \\ \vdots \\ \boldsymbol{\Psi}_{\partial_d} \end{bmatrix} \boldsymbol{c} \approx \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{y}_{\partial_1} \\ \vdots \\ \boldsymbol{y}_{\partial_d} \end{bmatrix}$$

Method justification

- This methodology works well because $(\sqrt{w}\,\psi_n'(x))$ also form an orthogonal system.

$\Rightarrow$ The columns of each matrix $\Psi, \Psi_{\partial_1} \ldots, \Psi_{\partial_n}$ are $\approx$ orthogonal as $N \to \infty$.

- Indeed, for instance, given two multi-indices $\boldsymbol{\alpha} \neq \boldsymbol{\beta}$, the product of their respective columns in Ψ_{∂_k} are

$$
\begin{aligned}
&\sum_{i=1}^{N} \frac{\partial}{\partial x_k}\psi_{\alpha}(\boldsymbol{x}^{(i)})\frac{\partial}{\partial x_k}\psi_{\beta}(\boldsymbol{x}^{(i)}) \\
&\quad = \sum_{i=1}^{N} \psi_{\alpha_1}(x_1^{(i)})\psi_{\beta_1}(x_1^{(i)}) \ldots \sqrt{w_k(x_k^{(i)})}\,\psi_{\alpha_k}'(x^{(i)})\sqrt{w_k(x_k^{(i)})}\,\psi_{\beta_k}'(x^{(i)}) \ldots \psi_{\alpha_d}(x_d^{(i)})\psi_{\beta_d}(x_d^{(i)}) \\
&\quad \approx N\mathbb{E}\left[\psi_{\alpha_1}(X_1)\psi_{\beta_1}(X_1) \ldots \sqrt{w_k(X_k)}\,\psi_{\alpha_k}'(X_k)\sqrt{w_k(X_k)}\,\psi_{\beta_k}'(X_k) \ldots \psi_{\alpha_k}(X_d)\psi_{\beta_k}(X_d)\right] \\
&\qquad = N\,\mathbb{E}\left[\psi_{\alpha_1}(X_1)\psi_{\beta_1}(X_1)\right] \ldots \mathbb{E}\left[w_k(X_k)\,\psi_{\alpha_k}'(X_k)\,\psi_{\beta_k}'(X_k)\right] \ldots \mathbb{E}\left[\psi_{\alpha_d}(X_d)\psi_{\beta_d}(X_d)\right] = 0.
\end{aligned}
$$

- Orthogonal polynomials, for example orthogonal polynomials, do not satisfy this property.

Sparse regression-based chaos expansions and Sobol' indices

We compute the coefficients $\boldsymbol{c}$ by ℓ^1-minimization:

$$\min_{\boldsymbol{c}} \left(\|\Psi \boldsymbol{c} - \boldsymbol{y}\|_2^2 + \sum_{k=1}^{d} \left\| \Psi_{\partial_k} \boldsymbol{c} - \boldsymbol{y}_{\partial_k} \right\|_2^2 + \gamma \|\boldsymbol{c}\|_1 \right).$$

Algorithm

- Least-angle regression (LARS) with λ chosen based on leave-one-out error (UQLab)

Sobol' indices

Computed as usual from the coefficients, e.g.

$$S_i^T = \frac{\sum_{\boldsymbol{\alpha} \in \mathcal{A}, \alpha_i > 0} c_{\boldsymbol{\alpha}}^2}{\sum_{\boldsymbol{\alpha} \in \mathcal{A}} c_{\boldsymbol{\alpha}}^2}$$

Global error

H^1 error on validation set:

$$E_{H^1} = \hat{\mathbb{E}}[(\mathcal{M}(\mathcal{X}_{\text{val}}) - \mathcal{M}_{\mathcal{A}}^{\text{surr}}(\mathcal{X}_{\text{val}}, \boldsymbol{c}))^2] + \sum_{k=1}^{d} \hat{\mathbb{E}} \left[\left(\frac{\partial \mathcal{M}}{\partial x_k}(\mathcal{X}_{\text{val}}) - \frac{\partial \mathcal{M}_{\mathcal{A}}^{\text{surr}}}{\partial x_k}(\mathcal{X}_{\text{val}}, \boldsymbol{c}) \right)^2 \right]$$

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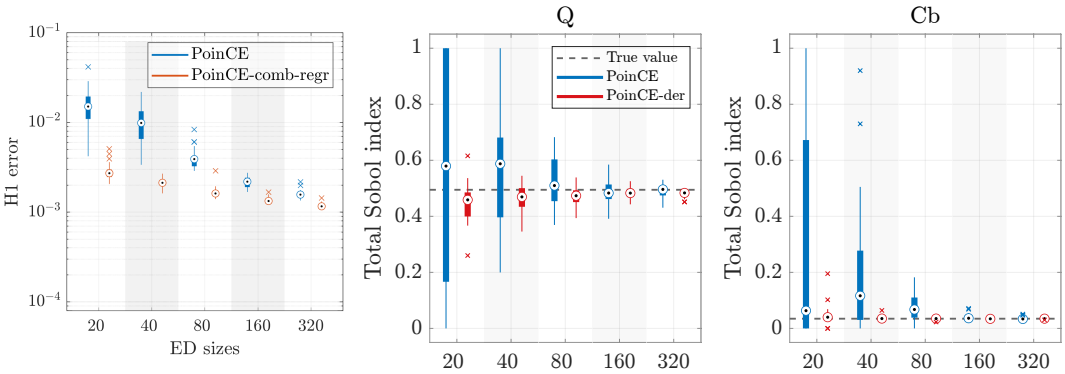
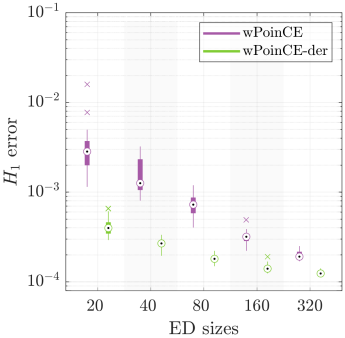
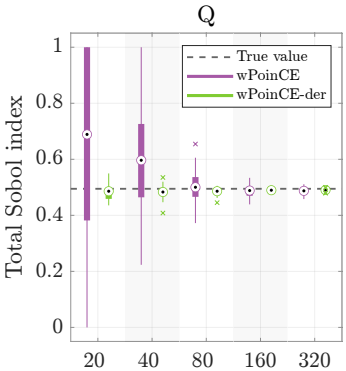
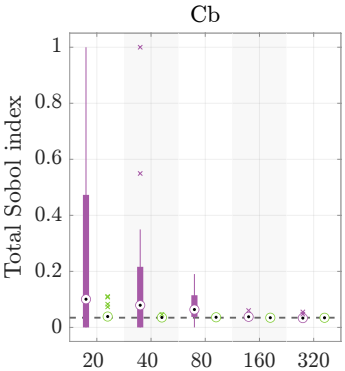
Flood cost model (unweighted)



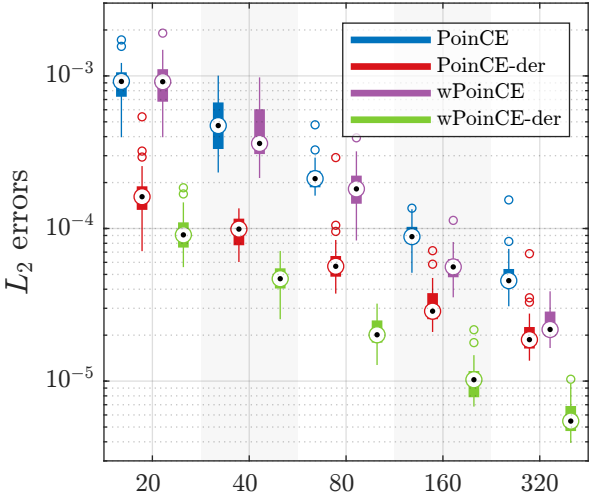
Flood cost model (weighted)







Flood cost model (unweighted vs weighted)



Literature

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